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An Identifiability-First Feasibility Protocol for a Gaussian-Regularized Diósi Collapse Test with a Casimir-Boundary Control

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Document type: theoretical design study and empirical-feasibility protocol

Version: referee-response draft, 30 July 2026

Authors and affiliations: to be supplied before external submission

Corresponding author: to be supplied before external submission

Scientific standing. This article does not report measured evidence for objective collapse, a physically realized mesoscopic superposition, a Casimir-to-collapse coupling, or spacetime-manifold dynamics. It reports a model-specific, synthetic identifiability result and defines the measurements required to determine whether that result survives in hardware.

Abstract

We present an identifiability-first protocol for testing one specified collapse dynamics: the nondissipative, Gaussian-regularized Diósi mass-density model applied to a single effective particle. Penrose objective reduction supplies motivation for asking whether spatially distinct mass-density branches have a finite lifetime; it does not supply the stochastic master equation used for the forecast. A nearby conducting boundary is treated as a secondary, independently controlled quantum-electrodynamic perturbation. No Casimir variable enters the registered collapse generator.

The methodological result is a design progression. An initial synthetic apparatus was rejected because its candidate collapse signature was nearly collinear with modeled nuisance responses: the maximum covariance-whitened signature cosine was 0.999977 and the normalized Gram condition number was 1.79×10^5 . A bounded synthetic redesign reduced those nominal values to 0.7177 and 6.53. Under that same assumed response and covariance model, the forecast power was 0.9979 and the nominal minimum was 542 paired windows. These power values are not empirical and are not treated as robust until a finite-pilot covariance and response-uncertainty campaign is completed.

The current reference manifest freezes a 276.3 nm-radius silica sphere with mass 1.94385×10^{-16} kg, branch separation 160 nm, 250 ms hold time, 1.2 μ m boundary gap, 4 K temperature, and 2×10^{-11} Pa pressure. At the selected regularization length $R_0=100$ nm, the registered effective-particle model predicts a conditional collapse rate of 0.0240 s^{-1} and a DP-only visibility loss of 0.598% at 250 ms. A separate 3.32% value belongs to a stronger sensitivity-grid cell and is not the reference-apparatus prediction.

The proposal remains an empirical no-go until the stated mass can be prepared in a verified spatial superposition, finite-geometry electromagnetic and material responses are measured, the covariance model passes preregistered stress tests, and an interpretation-eligible companion observable is demonstrated. The contribution is therefore a falsifiable protocol and a transparent map of its dominant failure modes, not evidence that objective collapse occurs.

Keywords: objective collapse; Diósi model; Penrose objective reduction; matter-wave coherence; Casimir boundary; identifiability; covariance whitening; experimental design

1. Scientific question and result

The experiment asks a narrow question:

After ordinary thermal, electromagnetic, gas, vibration, readout, and boundary-correlated responses have been measured, can the remaining complex coherence data support or exclude the frozen mass–separation–hold-time signature of one regularized Diósi model?

This question is intentionally narrower than “does gravity collapse the wave function?” A null result can constrain only the registered generator and parameter point within the experiment's demonstrated sensitivity. A positive residual can be assigned to that generator only if it is identifiable against the nuisance span and if the model's independently registered companion prediction is observed. Neither outcome, by itself, establishes Penrose's spacetime interpretation.

The present result is conditional:

1. a first synthetic design is non-identifiable and is rejected;
2. a bounded redesign is identifiable in its nominal synthetic world;
3. a single reference apparatus produces a 0.598% conditional DP-only coherence loss at 250 ms;
4. no measured response, covariance, state-preparation, or companion-detector receipt yet authorizes the acquisition-power forecast.

Figure 1 gives the proposed geometry. Figure 2 shows the acquisition sequence.

Current authoritative apparatus manifest

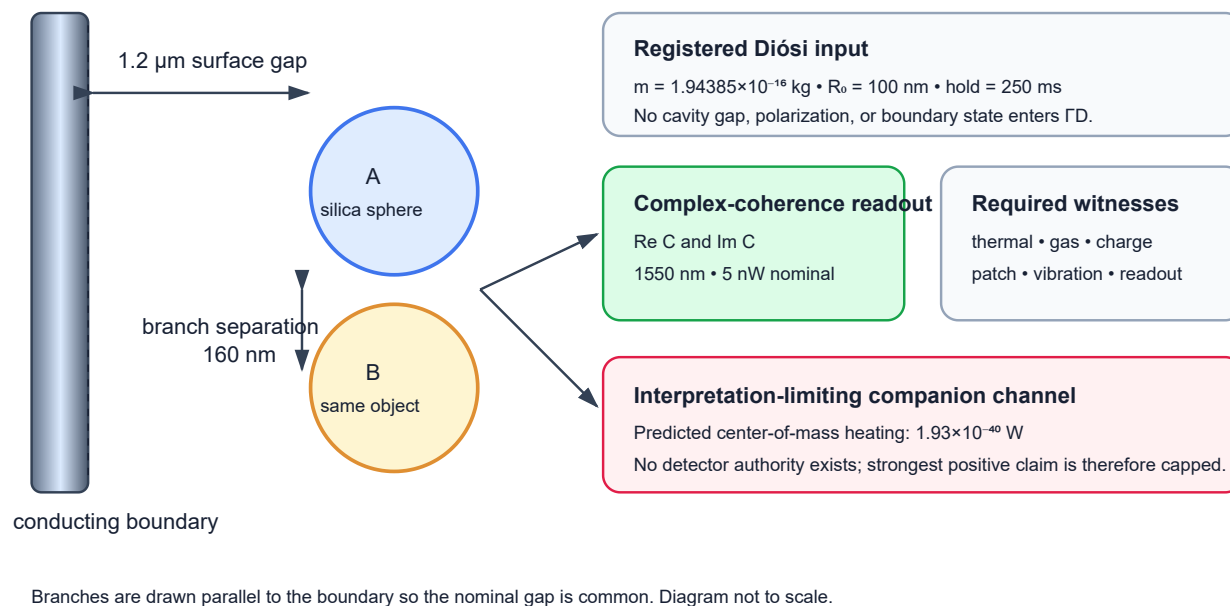
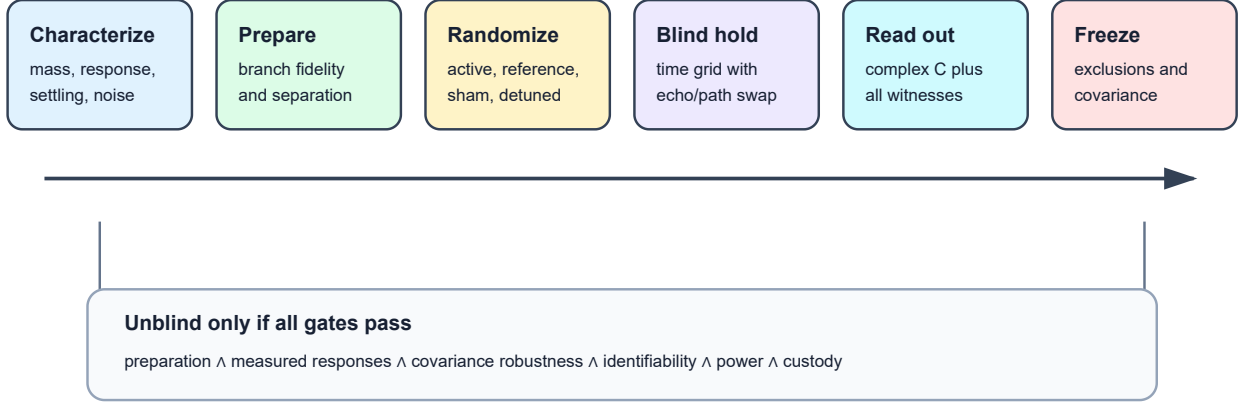


Figure 1. The boundary is an electromagnetic control, not a term in the registered Diósi generator. The two center-of-mass branches are separated parallel to the boundary so that the nominal surface distance is common to both branches. Environmental witnesses and a separate companion channel are required for interpretation.

Pilot and confirmatory timing logic



A fixed “0.5 Hz” label does not determine settling. Static randomized cells are the default until the measured transfer function authorizes a cadence.

Figure 2. Preparation and branch verification precede the blinded hold. Path swap, echo, sham-switch, and detuned-boundary cells are randomized. Unblinding is permitted only after exclusions, response vectors, covariance ancestry, and analysis code are frozen.

2. Exact tested dynamics

2.1 Registered Gaussian-regularized Diósi model

The executable prediction is not a generic “Diósi–Penrose” rate. It is one named reduced-order implementation: a nondissipative Diósi mass-density master equation for a single effective particle whose mass-density operator is Gaussian-smeared over the physical length R_0 [1–3]. In schematic form,

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}] - \frac{G}{2\hbar} \int d^3x d^3y \frac{[\hat{M}_{R_0}(\mathbf{x}), [\hat{M}_{R_0}(\mathbf{y}), \hat{\rho}]]}{|\mathbf{x} - \mathbf{y}|}. \quad (1)$$

The implementation freezes the convention, the mass representation, and R_0 . For two branches of the same effective Gaussian particle, with mass m and center separation d , the registered self-energy difference is

$$E_G(m, d, R_0) = Gm^2 \left[\frac{1}{\sqrt{\pi}R_0} - \frac{\text{erf}(d/2R_0)}{d} \right], \quad \Gamma_D = \frac{E_G}{\hbar}, \quad \mathcal{V}_D(t) = e^{-\Gamma_D t}. \quad (2)$$

The $d \rightarrow 0$ limit is evaluated analytically. A Fourier-space Simpson quadrature is used only as a numerical cross-check; its softening parameter is not the physical cutoff.

For the current reference point,

$$\begin{aligned}
m &= 1.94385 \times 10^{-16} \text{ kg}, \\
d &= 1.60 \times 10^{-7} \text{ m}, \\
R_0 &= 1.00 \times 10^{-7} \text{ m}, \\
E_G &= 2.53142 \times 10^{-36} \text{ J}, \\
\Gamma_D &= 2.40042 \times 10^{-2} \text{ s}^{-1}, \\
\tau_D &= 41.6594 \text{ s}.
\end{aligned} \tag{3}$$

At $t=0.25 \text{ s}$,

$$1 - \mathcal{V}_D = 1 - e^{-\Gamma_D t} = 5.98308 \times 10^{-3} = 0.598308\%. \tag{4}$$

This is the only headline coherence-loss forecast for the authoritative apparatus. The 3.32% loss reported in the development record belongs to the strongest transported sensitivity-grid cell, where $\Gamma_D = 0.134872 \text{ s}^{-1}$ at the same hold time. It is a different point, not a second estimate of Eq. (4).

2.2 Penrose OR motivation, notation, and scope

Penrose's objective-reduction argument motivates an order-of-magnitude lifetime $\tau_{\text{OR}} \sim \hbar/E_G$ for incompatible mass distributions [4,5]. That heuristic and Eq. (2) share a gravitational self-energy scale, but they are not the same theory object:

Question	Penrose OR motivation	Registered Diósi runtime
Why consider finite branch lifetime?	incompatibility of alternative spacetime geometries	not supplied
Numerical lifetime scale	$\tau \sim \hbar/E_G$	$\Gamma = E_G/\hbar$
Stochastic master equation	not uniquely specified	specified
Gaussian cutoff R_0	not fixed by OR	frozen at 100 nm
Diffusion/heating companion	not uniquely specified	specified
Casimir-boundary modifier	not supplied	absent

Consequently, an exclusion of Eq. (2) is not an exclusion of every Penrose-motivated theory. A match to Eq. (2) is not direct evidence for spacetime “snapping” or manifold instability.

2.3 Mass-density representation is part of the hypothesis

The reference object's geometric radius is 276.3 nm, but Eq. (2) treats its center of mass as one Gaussian-smeared effective particle. This is not equivalent to resolving the silica sphere's atomic or layered density. The current representation audit is:

Representation of the same apparatus	Current standing	Consequence
single effective Gaussian particle	executable; Eq. (2)	0.598% reference forecast
homogeneous rigid sphere	previous spatial calculation failed the registered convergence gate	no accepted comparative rate
coated or layered sphere	not computed	coating dependence unknown
coarse-grained internal density	not computed with admissible density provenance	granularity dependence unknown

Representation of the same apparatus	Current standing	Consequence
R0 sweep with converged representations	sensitivity-only calculations exist; no unified robustness receipt	parameter stability unresolved

Therefore this article proposes a test of the effective-particle Diósi model, not a representation-independent or generic DP test. A revised confirmatory analysis must either demonstrate stability across physically defensible mass representations or preserve that narrower claim in its title and conclusion.

3. Current Authoritative Apparatus Manifest

The following manifest supersedes all earlier candidate geometries in the development record.

Field	Authoritative value	Evidence role
identity	silica_high_mass_identifiable_single_object_v1	frozen design identity
material and geometry	silica sphere	design assumption
radius	276.302 nm	derived from frozen mass/material identity
mass	$1.94385 \times 10^{-16} \text{ kg} = 1.17061 \times 10^{11} \text{ Da}$	frozen model input
branch separation	160 nm, parallel to boundary	frozen model input
principal hold time	250 ms	frozen model input
hold-time requirement	include preregistered shorter and longer cells	identifiability check
nominal surface gap	1.2 μm , common to both branches	boundary-control input
boundary program	static randomized cells; nominal cadence label 0.5 Hz	transfer function unmeasured
primary sequence	Ramsey-type branch preparation/recombination	design assumption
control sequences	echo, path swap, sham switch, detuned boundary	required nuisance discrimination
temperature	4 K	design target, unmeasured in integrated apparatus
pressure	$2 \times 10^{-11} \text{ Pa}$	design target, unmeasured in integrated apparatus
vibration target	$5 \times 10^{-10} \text{ m s}^{-2} \text{ Hz}^{-1/2}$	design target
readout	1550 nm, 5 nW nominal	design assumption
polarization	circular-control pair plus linear-basis checks	electromagnetic witness
companion channel	center-of-mass energy increase	detector authority absent
settling rule	to be measured and frozen from transfer-function data	not inherited from old design

The “0.5 Hz” value is a configuration label, not permission to assume that a two-second cycle is quasistatic. No continuous modulation or ten-second settling rule is admitted until the measured boundary-to-apparatus transfer function defines a compatible sequence. Static randomized cells are the default pilot implementation.

3.1 Supersession ancestry

Design record	Geometry	Status
original symmetric-force candidate	early Casimir-force-centered architecture	superseded_design_record
proposal-closure architecture	75 nm radius, 20 nm separation, 0.1 s, 5 μm nominal gap, 10–4 μm commissioning ladder	superseded_design_record
synthetic identifiability candidate	high-mass parameter-search result	parent of current manifest
present reference manifest	276.3 nm radius, 160 nm separation, 0.25 s, 1.2 μm gap	current_authoritative_design_assumption

“Frozen” means immutable within a particular run. It does not mean that an older run remains the current proposal after an explicit downstream supersession.

3.2 The dominant scale gap

The reference mass is approximately 1.17×10^{11} Da. Matter-wave interference has recently been demonstrated for sodium clusters above 170,000 Da [6]. The proposed object is therefore about 6.89×10^5 times more massive than that free-particle interference benchmark. The comparison is not a universal impossibility theorem—trapped mechanical systems use different state-preparation strategies—but it prevents the manuscript from treating preparation as an ordinary checklist item.

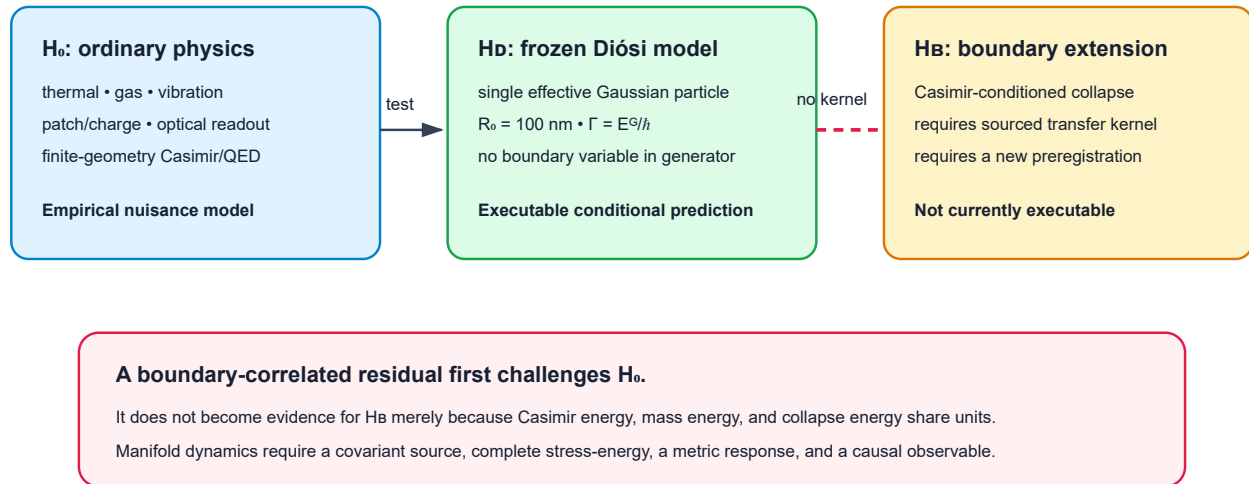
State preparation is the first physical go/no-go:

1. verify the object's mass and internal-state distribution;
2. demonstrate coherent branch preparation at the registered separation;
3. verify the 250 ms branch hold without conditioning on successful recombination;
4. measure visibility and phase stability over the full control schedule;
5. show that the preparation procedure does not itself generate the fitted nuisance signature.

Until these steps are demonstrated, neither 542 nor 1,600 paired windows is an actionable acquisition duration.

4. Hypotheses separated before data

Three hypotheses; no silent bridge



Penrose OR motivates the question but does not uniquely supply H₀ or H_B.

Figure 3. The ordinary-physics null, the frozen effective-particle Diósi generator, and a possible boundary-to-collapse extension are separate hypotheses. A Casimir-correlated residual cannot move from the left branch to the right branch without a registered transfer kernel and new data.

4.1 Ordinary-physics null, H₀

H₀ contains all registered ordinary mechanisms that can alter complex coherence:

- blackbody emission, absorption, scattering, and temperature drift;
- residual-gas collision and recoil;
- patch potentials, charge, dielectric response, and electromagnetic forces;
- vibration, tilt, support motion, and clock error;
- optical back-action, detector self-noise, and analysis leakage;
- boundary switching, material hysteresis, and finite-geometry Casimir/QED response.

The nuisance model is empirical. Textbook formulas provide priors and dimensional checks, not a substitute for measured response vectors.

4.2 Frozen Diósi hypothesis, H_D

H_D adds Eq. (2) to H₀ with the mass, separation, R_0 , and hold-time law frozen before confirmatory data. The Casimir gap and modulation state do not enter the collapse generator. Under complete joint-system equivalence, standard H_D therefore predicts the same collapse term in boundary-active and boundary-reference cells.

4.3 Lane C: boundary-conditioned extension, HB

HB is a future model slot in which a Casimir boundary changes collapse or coherence through an explicitly sourced transfer kernel. No such kernel is registered. The ideal Casimir energy,

$$\frac{E_C}{A} = -\frac{\pi^2 \hbar c}{720 a^3}, \quad P_C = -\frac{\pi^2 \hbar c}{240 a^4}, \quad (5)$$

does not define a map from plate gap a to Γ_D . Real-material Lifshitz theory, finite geometry, temperature, patches, roughness, and nonequilibrium response belong first in H0.

4.4 Manifold-response hypothesis

The motivating manifold statement is a research question, not an admitted mechanism: if alternative mass configurations source incompatible gravitational geometries, a collapse-like lifetime might scale with a gravitational self-energy. The present nonrelativistic master equation tests one phenomenological realization of that scale. A spacetime claim would additionally require a covariant source, a complete apparatus stress-energy tensor, a specified dynamical metric response, and a causal observable. Those objects are absent.

4.5 Compton-frequency non-bridge

Rest energy may be written as a frequency, $\nu_C = mc^2/h$, and the collapse energy as $\nu_G = E_G/h$. Atomic, QED, Higgs, blackbody, Zeeman, Stark, Maxwell, and gravitational benchmarks use compatible energy and frequency units. That compatibility is valuable for calibration and unit recovery but does not create a coupling between cavity modes and collapse:

$$\mathcal{K}_{\text{boundary} \rightarrow \text{branch/coherence}} \quad \text{is not registered.} \quad (6)$$

The same rule applies to light cones, spinors, gravitational waves, Jeans instability, Schwarzschild radii, and mass–energy equivalence. They validate parts of relativistic or quantum notation; they do not supply Eq. (6).

5. Complex-coherence estimator and identifiability

The primary datum is not a fitted scalar decay constant. Each control cell contributes a complex coherence,

$$C_k = \langle e^{i\phi} \rangle_k = \text{Re } C_k + i \text{Im } C_k. \quad (7)$$

Real and imaginary components retain phase rotations that a visibility-only analysis would discard. Let y be the stacked real vector of all components, X the matrix of registered nuisance signatures, s_D the frozen Diósi signature, and Σ the block covariance. Whitening gives

$$\tilde{y} = L^{-1}y, \quad \tilde{X} = L^{-1}X, \quad \tilde{s}_D = L^{-1}s_D, \quad LL^T = \Sigma. \quad (8)$$

The collapse-sensitive component is the part orthogonal to the whitened nuisance span:

$$s_{\perp} = (I - P_{\tilde{X}})\tilde{s}_D. \quad (9)$$

The preregistered nominal design gates are

$$\max_j |\cos(\tilde{s}_D, \tilde{x}_j)| \leq 0.97, \quad \kappa(G_{\text{norm}}) \leq 100, \quad \text{power} \geq 0.80, \quad \text{FPR} \leq 0.05. \quad (10)$$

These are design-admission gates, not proof that the modeled nuisance set is complete.

5.1 Rejected design

The initial synthetic design produced

Diagnostic	Result	Gate
maximum absolute whitened cosine	0.999977	fail
normalized Gram condition number	179,104	fail
acquisition power	withheld	not meaningful

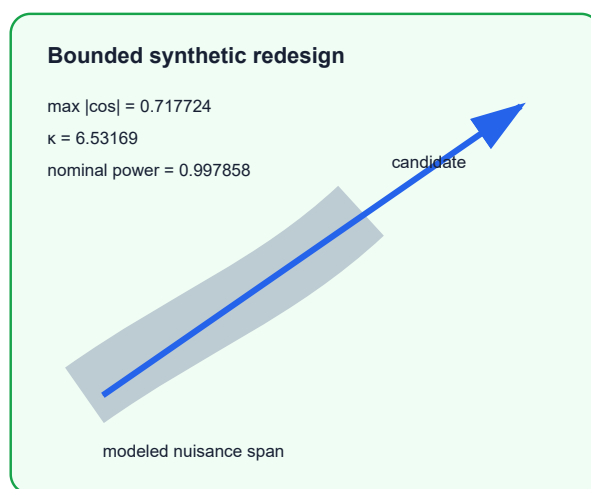
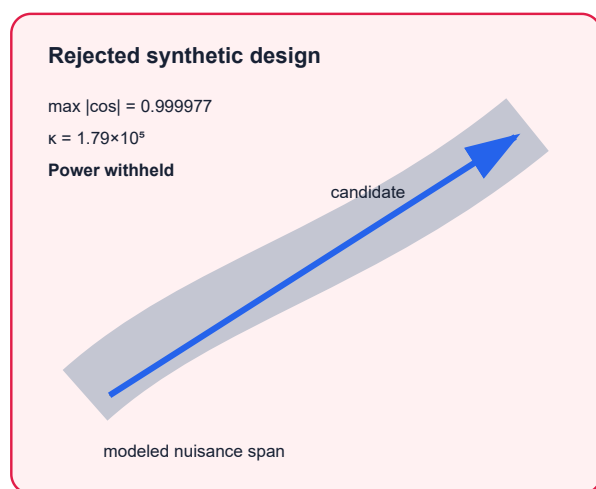
The correct result was an apparatus-design no-go. It was not a null result on the Diósi model.

5.2 Nominal synthetic redesign

The bounded redesign produced

Diagnostic	Nominal synthetic value	Gate
maximum absolute whitened cosine	0.717724	pass
normalized Gram condition number	6.53169	pass
forecast power	0.997858	pass in nominal world
nominal required paired windows	542	descriptive only
planned paired windows	1,600	not yet authorized

Covariance-whitened identifiability



The right panel is not empirical. Finite-pilot covariance, response error, missing channels, drift, tails, and selection optimism remain unstressed.

Figure 4. The first design's candidate direction lies almost inside the nuisance span. The redesign is separated in the nominal whitened geometry. The right panel is still conditional on synthetic response vectors and an assumed covariance.

5.3 Why the nominal power is not the headline

The same synthetic response/covariance world was used to search candidates and evaluate the selected candidate. Finite-pilot covariance uncertainty was not propagated. Accordingly, 0.997858 is a conditional calculation, not a robust power claim. The next computation must report an envelope rather than a single number.

The preregistered stress family must include:

Stress axis	Required analysis	Current status
finite-pilot covariance	nested simulation, bootstrap, or analytical propagation	not run
shrinkage/regularization	repeat whitening across declared estimators	not run
drift and non-Gaussian tails	correlated drift and heavy-tail worlds	not run
response amplitude/phase error	perturb every nuisance vector within pilot uncertainty	not run
missing nuisance	inject unregistered channels and test residual inflation	not run
branch and hold-time jitter	propagate preparation/metrology distributions	not run
shared calibration ancestry	model cross-control correlation explicitly	partially represented, not stressed
leave-one-control-out	refit after withholding each control family	not run
covariance structure	alternate preregistered block families	not run
selection optimism	repeat candidate search in independent synthetic worlds	not run

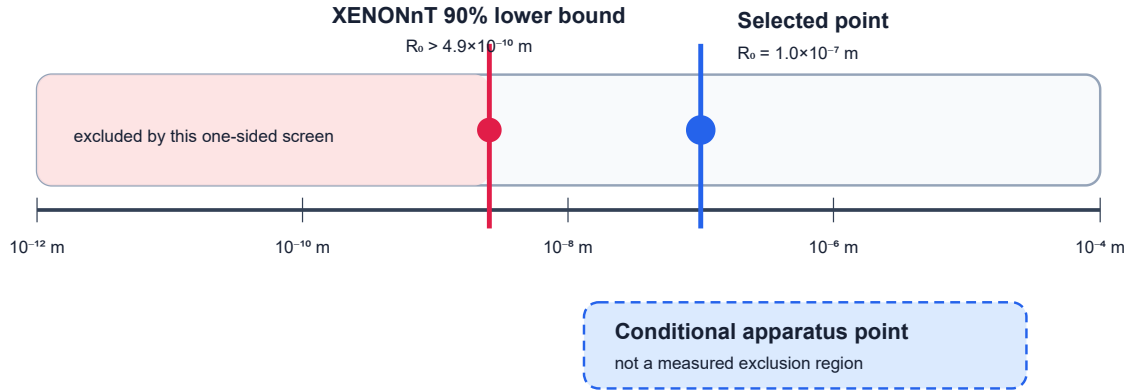
The accepted future headline has the form: “Across the registered uncertainty family, power ranged from X to Y and all design gates survived in Z% of worlds.” Until X, Y, and Z are computed, the acquisition-power claim remains provisional.

6. External constraints and scientific reach

The selected $R_0=100$ nm point must be compared with constraints on the same model convention. Direct spontaneous-radiation searches currently provide a strong lower-bound axis for Markovian DP. The 2026 XENONnT analysis reports $R_0 > 4.9 \times 10^{-10}$ m at 90% confidence and $R_0 > 4.5 \times 10^{-10}$ m at 95% confidence in its stated Diósi convention [7]. Earlier germanium searches ruled out the natural parameter-free proposal and progressively strengthened the lower limit [8,9].

The selected $R_0 = 1.0 \times 10^{-7}$ m is about 204 times the XENONnT 90%-confidence lower bound. It is therefore not excluded by that one-sided screening comparison. This is not yet a full admission receipt: the mass-density smearing definition, normalization convention, charged-constituent radiation map, and treatment of the effective composite particle must be matched line by line before the point is called externally allowed.

Regularization-length screening (model mapping still required)



The axis comparison is numerical only until the Gaussian convention, normalization, composite-particle mapping, and radiation response are matched.

Figure 5. Logarithmic R_0 screening. The XENONnT lower bound and the selected 100 nm point are shown on the same numerical axis. The shaded “apparatus” marker is conditional sensitivity, not a measured exclusion. The formal model-equivalence audit remains open.

The current reach statement is therefore limited:

- a measured null with validated sensitivity could exclude the registered effective-particle point;
- it would not exclude all regularized, dissipative, colored-noise, or Penrose-motivated models;
- a positive coherence residual could favor the frozen line shape over the registered nuisance span;
- it could not distinguish neighboring collapse generators without additional parameter and companion observables;
- a boundary-correlated excess would first challenge H_0 and the joint-system equivalence assumption, not prove HB.

7. Final reference-apparatus forecast

Figure 6 shows the conditional coherence curve from Eq. (2). It contains no ordinary-decoherence band because that band must be learned from pilot data.

Conditional model-only visibility loss

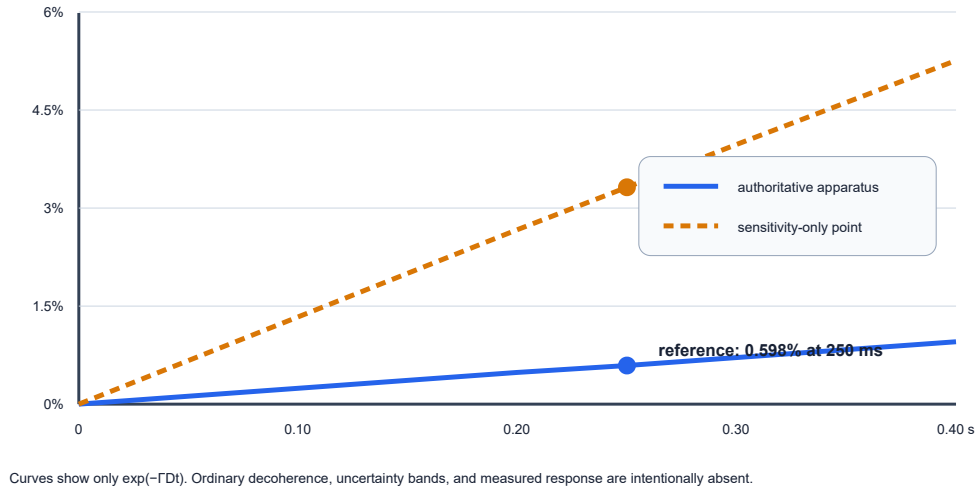


Figure 6. The authoritative reference curve reaches a 0.598% loss at 250 ms. The sensitivity-grid curve reaches 3.32% and is shown only to prevent those predictions from being conflated. Neither curve is a total visibility forecast.

7.1 Companion observable and interpretation ceiling

The same nondissipative generator predicts momentum diffusion

$$D_{pp} = \frac{G\hbar m^2}{12\sqrt{\pi}R_0^3} \quad (11)$$

and a center-of-mass energy increase

$$\dot{E} = \frac{3D_{pp}}{m} = \frac{G\hbar m}{4\sqrt{\pi}R_0^3}. \quad (12)$$

For the reference manifest, $\dot{E} = 1.92979 \times 10^{-40}$ W. If 100 independent samples and $\text{SNR} \geq 5$ were demanded, the algebraic maximum one-shot standard uncertainty would be

$$\sigma_{\dot{E}, \text{one shot}} \leq \frac{\dot{E}\sqrt{100}}{5} = 3.85958 \times 10^{-40} \text{ W}. \quad (13)$$

Equation (13) is not an instrument model. No registered detector demonstrates this bandwidth, calibration, independence, or noise floor. The companion observable is therefore presently a structural no-go for the paper's strongest positive interpretation.

This revision adopts the conservative third option:

Unless an independently powered companion channel becomes feasible, a replicated DP-shaped coherence residual may be reported as unexplained model-consistent phenomenology, but not as support-eligible identification of the Diósi generator.

The project may replace the heating channel only through a preregistered derivation from the same frozen dynamics, with an instrument-level bandwidth, integration-time, noise, calibration-transfer, and cross-covariance model.

8. Empirical pilot and confirmatory decisions

8.1 Pilot first

The feasibility pilot is a hardware-characterization campaign, not a search for collapse. It must deliver provenance-bound measurements of:

1. object mass, geometry, material, coating, and internal-density model;
2. state-preparation fidelity and branch separation;
3. hold-time and recombination metrology;
4. finite-geometry electromagnetic Green response and material permittivity;
5. boundary-switching transfer functions and settling time;
6. thermal, gas, vibration, tilt, patch, charge, optical, and sensor responses;
7. raw complex-coherence covariance, including shared calibration ancestry;
8. sham, detuned, echo, and path-swap controls;
9. companion-channel feasibility or the lower interpretation ceiling;
10. blinded data custody and independent solver replication.

8.2 Pilot admission rule

$$\mathcal{G}_{\text{pilot}} = \mathcal{G}_{\text{preparation}} \wedge \mathcal{G}_{\text{response}} \wedge \mathcal{G}_{\text{covariance}} \wedge \mathcal{G}_{\text{identifiability}} \wedge \mathcal{G}_{\text{power}} \wedge \mathcal{G}_{\text{custody}}. \quad (14)$$

The pilot is admitted to confirmatory planning only if every factor is true. The numerical identifiability gates remain Eq. (10), but power must additionally survive the uncertainty envelope in Section 5.3. If any factor fails, the result is an explicit apparatus-redesign no-go.

8.3 Confirmatory campaign

Only after pilot admission may the team freeze:

- one canonical apparatus identity;
- one mass-density representation and R0 point or preregistered grid;
- exclusions and quality-control rules;
- response and covariance ancestry;
- analysis code and randomization;
- companion interpretation ceiling;
- primary and replication sample sizes.

The confirmatory data remain blinded until completeness and custody checks pass. An independent replication is required for any positive interpretation.

8.4 Decisive outcomes

Observed outcome	Defensible inference	Not established
pilot cannot prepare the state	current apparatus infeasible	collapse model false
pilot fails cosine/conditioning gates	current design non-identifiable	collapse model false
measured null with demonstrated sensitivity	registered point disfavored/excluded at stated confidence	all DP or Penrose OR false
residual follows ordinary control	H0 response model repaired	collapse
boundary-correlated residual without transfer kernel	anomaly eligible for a new bridge study	Casimir-induced collapse
replicated Diósi-shaped coherence residual, no companion	unexplained model-consistent phenomenology	generator identified
replicated shape plus independent companion	tested generator gains support within registered alternatives	manifold dynamics or universal objective collapse

9. Observable-separation gate

Force, phase, coherence, heating, curvature, and collapse rate are different observables. They may be connected only by a declared model with units, parameters, provenance, and falsifiers. The governing gate is:

$$\text{observable A} \rightarrow \text{observable B} \implies \{\text{transfer law, units, source, parameters, test}\}. \quad (15)$$

Equation similarity, shared use of Planck's constant, or a common energy-frequency conversion is insufficient. In particular:

- Casimir pressure is not DP self-energy;
- renormalized negative energy density is not automatically negative spacetime curvature for the complete apparatus;
- virtual-particle language is not a detector model;
- gravitational-wave interference does not imply phase coherence between Casimir and matter-wave channels;
- circular polarization is an electromagnetic control degree of freedom, not a collapse source;
- a light cone defines causal structure, not a nonrelativistic collapse generator.

10. Limitations

The main limitations are scientific, not typographic:

1. **State preparation:** the reference mass is nearly 690,000 times the 170 kDa free-particle interference benchmark.
2. **Synthetic optimization:** the selected design was evaluated in the same assumed response/covariance world in which it was found.
3. **External-bound mapping:** the selected R0 passes a numerical lower-bound screen, but the exact convention/composite mapping is not certified.

4. **Mass representation:** the 0.598% prediction is specific to one effective Gaussian particle; rigid, layered, and coarse-grained comparisons are incomplete.
5. **Companion channel:** the heating signal lacks a physically credible detector receipt, lowering the maximum positive interpretation.
6. **Ordinary physics:** finite-geometry material and environmental responses are unmeasured for the integrated apparatus.
7. **Relativity:** the registered dynamics is nonrelativistic and does not compute manifold evolution.
8. **Boundary bridge:** no Casimir-to-collapse transfer kernel exists in the tested model.

The machine-readable status is equivalent to the following prose: measured evidence is not yet available; empirical pilot readiness and physical viability have not been established; collapse identification and manifold dynamics remain blocked.

11. Discussion

11.1 What is new

The useful result is not that a sufficiently massive object has a large collapse rate; that scaling is already built into the selected model. The advance is the conversion of that rate into a fail-closed experimental comparison. A candidate signal is expressed in the same complex-coherence coordinates as the measured control responses, whitened by their joint covariance, and tested for linear dependence before acquisition power is reported. The rejected design demonstrates why this order matters. A nominally nonzero loss can be statistically powerless as evidence when its direction is indistinguishable from ordinary drift or decoherence.

This approach also changes what counts as a useful negative result. Failure to prepare the state is an apparatus no-go. Failure of the cosine or conditioning gate is an identifiability no-go. A measured null becomes a model constraint only after the apparatus demonstrates sensitivity to the frozen direction. These outcomes answer different questions and should not be combined into a single “experiment failed” category.

11.2 Why retain the Casimir boundary

The Casimir boundary remains valuable even though it does not enter Eq. (2). It creates a tunable electromagnetic and material environment with strong gap, temperature, polarization, and surface dependence. That makes it an unusually demanding test of whether the ordinary-physics null is complete. Sham switching, detuning, static gap cells, path swaps, and polarization controls can reveal phase rotations or losses that might otherwise be misidentified as a mass-dependent collapse residual.

This use is scientifically asymmetric. Agreement with finite-geometry macroscopic QED improves confidence in H_0 ; disagreement produces a boundary-response anomaly. Neither result modifies HD automatically. Only a new, independently motivated transfer law could make the boundary a parameter of a collapse generator. Keeping that bridge closed protects both sides of the experiment: ordinary boundary physics is not mislabeled as gravity, and a failure of the particular Diósi model is not mislabeled as a failure of Casimir theory.

11.3 What a positive result would mean

A coherence residual with the registered mass, separation, and hold-time dependence would be more informative than an unexplained scalar decay rate. Replication across object masses and branch geometries would make an ordinary single-channel explanation progressively less plausible. Even so, the interpretation remains conditional on the alternative set. An unmodeled gas, patch, readout, preparation, or calibration response can imitate a structured signal if it shares the same experimental schedule.

The companion channel was intended to reduce that ambiguity by demanding a second prediction from the same generator. Its forecast, however, is too small to count as an experimental discriminator without a credible instrument proposal. The present claim ceiling is therefore deliberately lower than the original protocol's aspirational ceiling. A replicated coherence residual without a companion would motivate new tests and model comparison; it would not identify objective collapse. This is a scientific consequence of the detector gap, not merely cautious wording.

11.4 What a null result would mean

At a demonstrated sensitivity, a null result can exclude the registered effective-particle point. It cannot directly exclude a rigid-sphere representation that was never converged, a different regularization length, a dissipative or colored-noise theory, or Penrose's broader lifetime motivation. The mass-density robustness campaign therefore determines how widely the null may be interpreted. If several defensible representations predict comparable losses and all are within reach, the experiment tests a family. If their predictions diverge substantially, the representation itself becomes part of the preregistered alternative set.

11.5 Priority order after this design study

The next scientific work should be ordered by the chance that it invalidates the apparatus before costly acquisition:

1. demonstrate or reject state preparation at the registered mass, separation, and hold time;
2. complete the exact external-bound convention and composite-particle map;
3. measure finite-geometry material and boundary transfer functions;
4. obtain pilot response vectors and covariance;
5. run the registered robustness envelope and repeat candidate selection in independent synthetic worlds;
6. compare mass-density representations on a converged numerical grid;
7. either demonstrate a companion instrument or retain the lower interpretation ceiling;
8. only then freeze a blinded confirmatory campaign.

This sequence keeps the paper's methodological core useful even if the present apparatus is rejected. The estimator, hypothesis separation, packet custody, and design gates can be carried into a lower-mass or otherwise redesigned platform without preserving the current sphere as a favored physical object.

12. Claim boundaries

This paper establishes:

- the exact model and parameter point used by the forecast;

- the failure of the first synthetic identifiability design;
- the nominal separability of a bounded redesign under stated assumptions;
- a single, unambiguous 0.598% reference-apparatus prediction;
- a fail-closed pilot and confirmatory decision structure;
- the empirical and computational work still required.

This paper does not establish:

- that the apparatus can be built or the state prepared;
- that the nominal power survives model uncertainty;
- that the selected point is fully admitted by all external constraints;
- that a Casimir boundary changes a collapse rate;
- that objective collapse occurs;
- that Penrose's geometric interpretation is observed;
- that spacetime manifold dynamics have been measured or simulated.

13. Conclusion

The strongest result is methodological. A scalar loss rate that is nearly collinear with ordinary backgrounds is not an adequate collapse test. The experiment must operate in raw complex-coherence space, measure the nuisance responses and covariance, project the frozen candidate away from that span, and fail closed when the geometry is ill-conditioned.

The current synthetic redesign satisfies nominal identifiability gates and maps one effective-particle Diósi model to a 0.598% loss at 250 ms. That result is scientifically useful because it identifies exactly what hardware would need to resolve. It is not yet a physically viable proposal: the state-preparation scale, response/covariance robustness, exact external-bound mapping, mass-density dependence, and companion detector remain open.

The appropriate next action is therefore a measured feasibility pilot. If the pilot reproduces the response geometry and survives the registered robustness family, a blinded confirmatory campaign becomes justified. If it does not, the apparatus is redesigned or rejected without converting an engineering failure into a claim about nature.

Publication declarations

Author contributions. To be completed by the human authors before submission. Repository history records the provenance of calculations and AI-assisted drafting; it does not determine scholarly authorship.

Code and data availability. The executable configurations, synthetic fixtures, reports, tests, equation-action sidecars, and receipts are maintained in the CasimirBot repository. The complete development record is preserved in [the reproducibility supplement](#). No measured experimental dataset is reported.

Competing interests. To be declared by the human authors before submission.

Acknowledgments. To be completed before submission.

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